

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(lubridate)
library(rvest)
library(here)
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#create my theme
mytheme <- theme_classic(base_size = 12)
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
```

```
## List of 2
##  $ axis.text      :List of 11
##   ..$ family      : NULL
##   ..$ face        : NULL
##   ..$ colour      : chr "black"
##   ..$ size        : NULL
```

```
## ..$ hjust      : NULL
## ..$ vjust      : NULL
## ..$ angle      : NULL
## ..$ lineheight : NULL
## ..$ margin     : NULL
## ..$ debug      : NULL
## ..$ inherit.blank: logi FALSE
## ..- attr(*, "class")= chr [1:2] "element_text" "element"
## $ legend.position: chr "top"
## - attr(*, "class")= chr [1:2] "theme" "gg"
## - attr(*, "complete")= logi FALSE
## - attr(*, "validate")= logi TRUE
```

```
#set mytheme to standard theme
theme_set(mytheme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
LWSP_website <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```

#3
water_system_name <- LWSP_website %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()
PWSID <- LWSP_website %>%
  html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()
Ownership <- LWSP_website %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()
MaxDayUse <- LWSP_website %>%
  html_nodes('th~ td+ td , th~ td+ td') %>%
  html_text()

```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

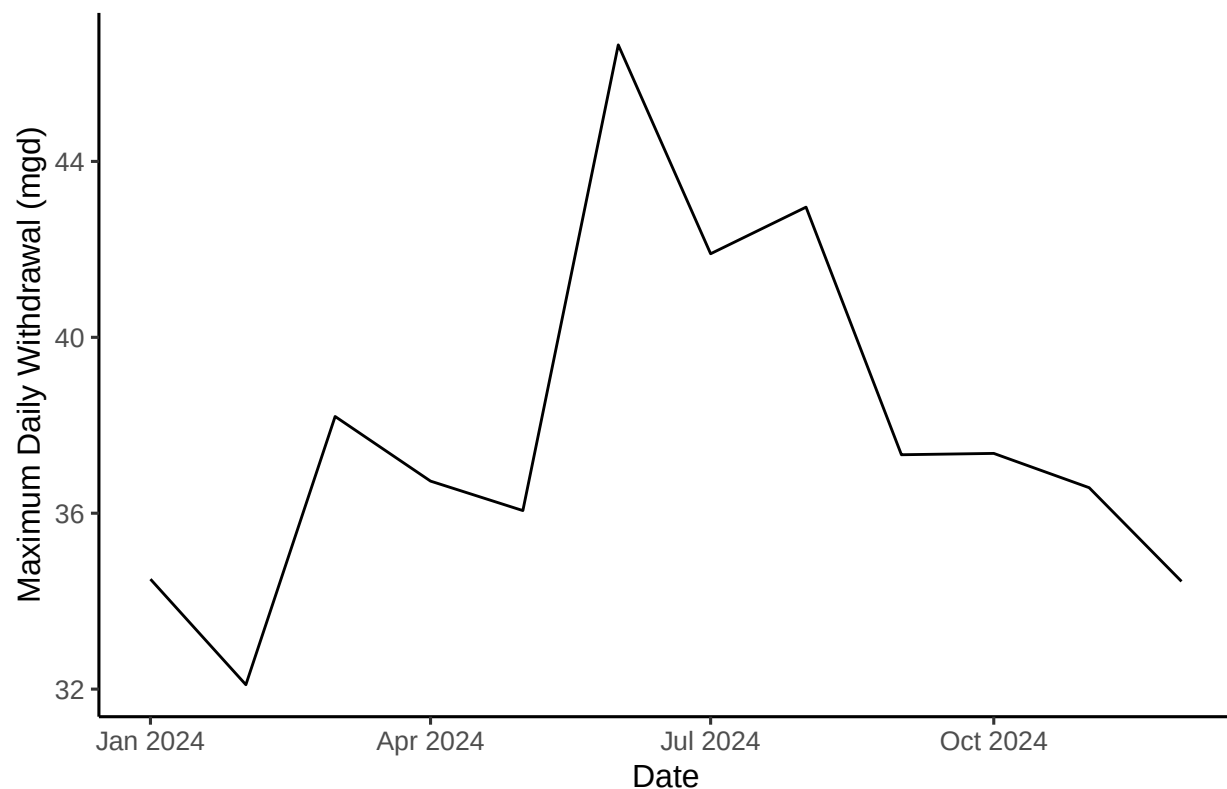
```

#4
water_df <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                      "Year" = rep(2024, 12),
                      "MaxDayUse" = as.numeric(MaxDayUse)) %>%
  mutate(
    Water_System_Name = !!Ownership,
    PWSID = !!PWSID,
    Ownership = !!Ownership,
    Date = my(paste(Month, "-", Year)))

#5
ggplot(water_df, aes(x=Date,
                    y = MaxDayUse))+
  geom_line()+
  labs(
    title = "2024 Durham Muncipal Water Supply Maximum Daily Withdrawal",
    y = "Maximum Daily Withdrawal (mgd)",
    x = "Date")

```

2024 Durham Municipal Water Supply Maximum Daily Withdrawal



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

#6.

```
##LWSP_website <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')

## the_base_url <- 'https://www.ncwater.org/WUDC/app/LWSP/report.php?'
## the_facility <- '03-32-010'
## the_year <- 2024
## the_scrape_url <- paste0(the_base_url,'pwsid=',the_facility,'&year=', the_year)
## print(the_scrape_url)

scrape.it <- function(the_facility, the_year){
  #get proper URL
  the_base_url <- 'https://www.ncwater.org/WUDC/app/LWSP/report.php?'
  the_url <- paste0(the_base_url,'pwsid=',the_facility,'&year=', the_year)

  #get the website
```

```

LWSP_website <- read_html(the_url)
print(the_url)

#scrape the data
water_system_name <- LWSP_website %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()
PWSID <- LWSP_website %>%
  html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()
Ownership <- LWSP_website %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()
MaxDayUse <- LWSP_website %>%
  html_nodes('th~ td+ td , th~ td+ td') %>%
  html_text()

#convert the the dataframe
water_df <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                      "Year" = rep(the_year, 12),
                      "MaxDayUse" = as.numeric(MaxDayUse)) %>%
  mutate(
    Water_System_Name = !!Ownership,
    PWSID = !!PWSID,
    Ownership = !!Ownership,
    Date = my(paste(Month, "-", Year)))

#return the dataframe
return(water_df)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7
durham_water_2020 <- scrape.it('03-32-010',2020)

```

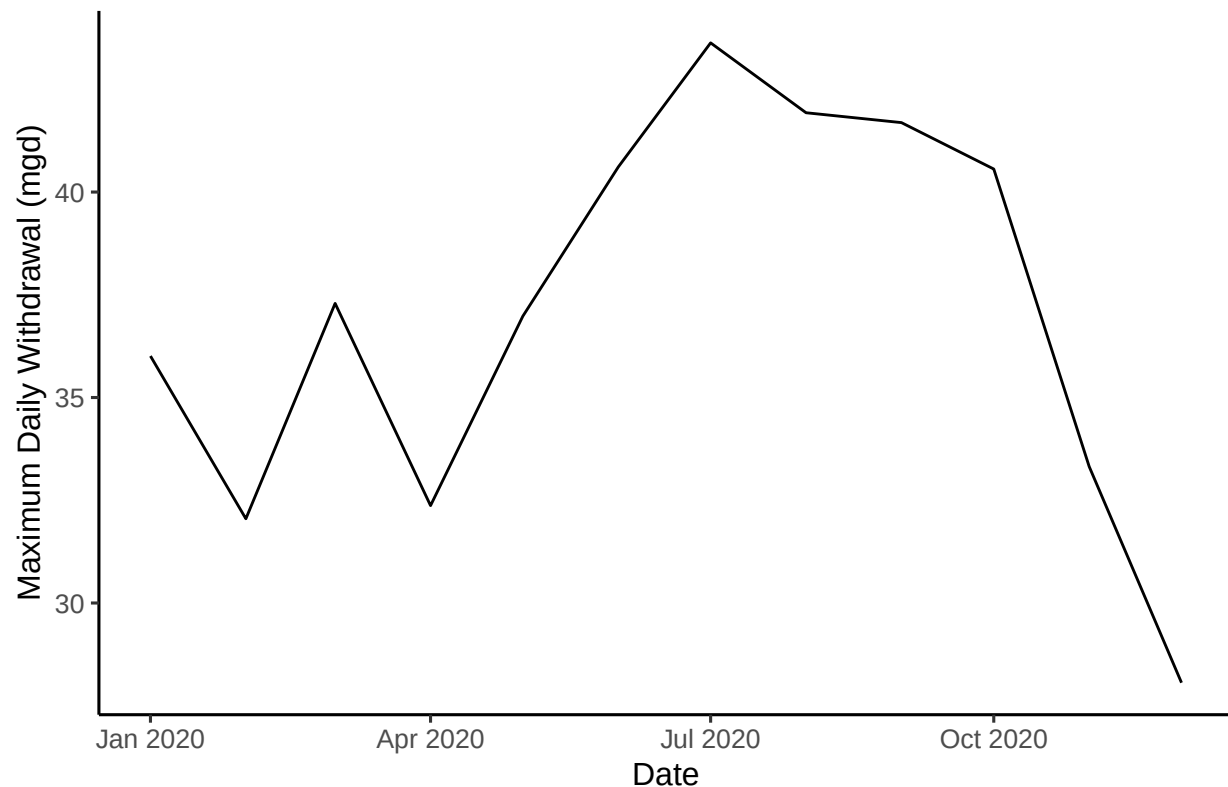
```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2020"
```

```

#graph
ggplot(durham_water_2020, aes(x=Date,
                             y = MaxDayUse))+
  geom_line()+
  labs(
    title = "2020 Durham Muncipal Water Supply Maximum Daily Withdrawal",
    y = "Maximum Daily Withdrawal (mgd)",
    x = "Date")

```

2020 Durham Municipal Water Supply Maximum Daily Withdrawal



```
#show structure
glimpse(durham_water_2020)
```

```
## Rows: 12
## Columns: 7
## $ Month      <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Year       <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020
## $ MaxDayUse  <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 43.~
## $ Water_System_Name <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ PWSID      <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ Ownership  <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Date       <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, 2020~
```

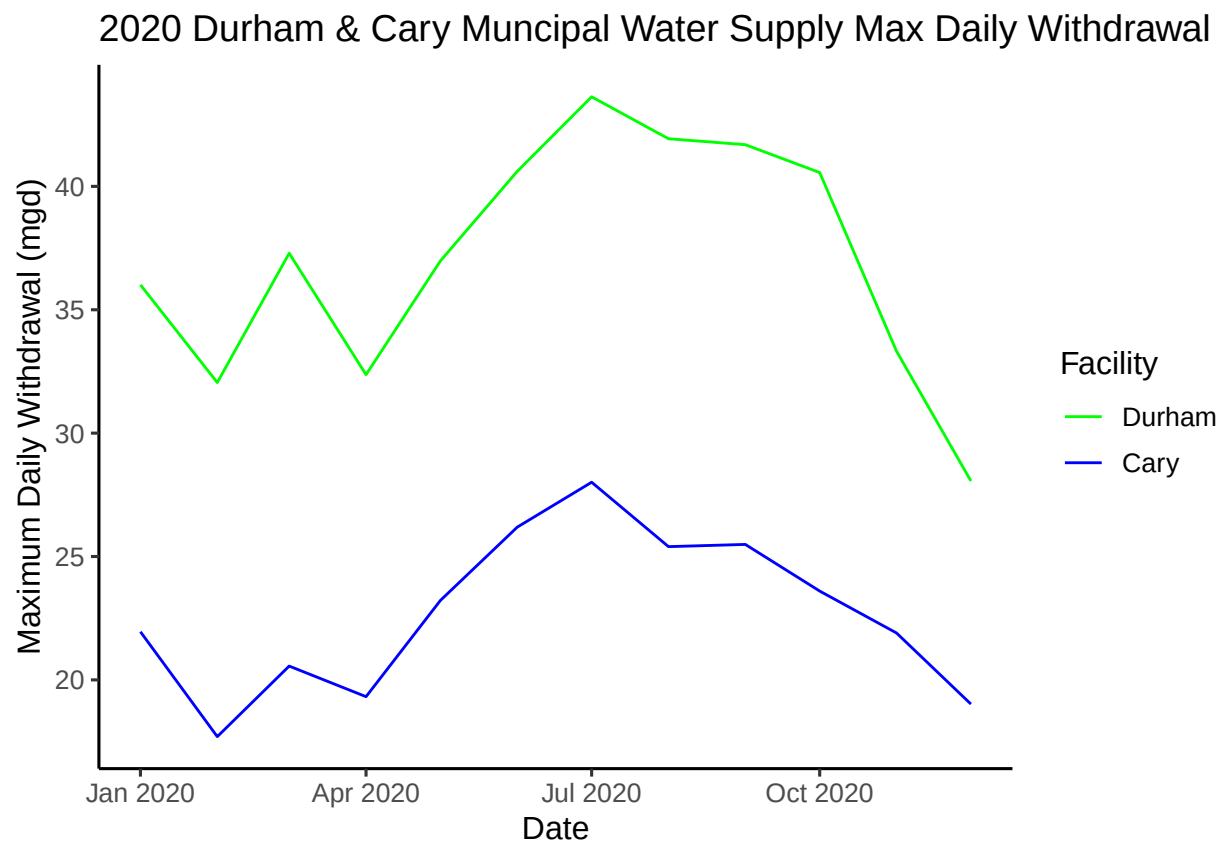
- Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
#get cary 2020 water data
cary_water_2020 <- scrape.it('03-92-020',2020)
```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2020"
```

```
#combine with Durham data
cary_durham_water <- rbind(durham_water_2020, cary_water_2020)

#graph
ggplot(cary_durham_water, aes(x = Date, y = MaxDayUse, color = PWSID))+
  geom_line()+
  labs(
    title = "2020 Durham & Cary Muncipal Water Supply Max Daily Withdrawal",
    y = "Maximum Daily Withdrawal (mgd)",
    x = "Date",
    color = "Facility")+
  scale_color_manual(values = c('green', 'blue'), labels = c('Durham', 'Cary'))
```



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

#9

Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: >